

SANKUR KUNDU

Software Engineer | Backend, Systems & Full-Stack

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SUMMARY

Software engineer (B.Tech CSE, 2026) who builds systems from scratch and proves them correct: **relay** (a fault-tolerant LLM inference cluster in **C++/CUDA** and **Swift**), **tandem** (distributed training **bit-identical to PyTorch**), **quorumbd** (TLA+-verified SQL), **tessel** (GPU kernels at **cuBLAS parity**). Co-founded Gigzen as its sole engineer.

PROJECTS

relay — C++, CUDA and Swift: a disaggregated, fault-tolerant LLM inference cluster

- Built **C++/CUDA** prefill/decode workers with my own **flash-attention** and **flash-decoding** kernels, a **KV transfer engine** streaming each layer as the next computes, and a **Swift** control plane (**C++ interop**, OpenAI API, **prefix-aware routing**).
- On a **T4**, my kernels cut 2,048-token prefill **8.7× (3.7 s → 0.43 s)** and batch-32 decode **5.4×**; streaming hides **99–100%** of the KV transfer; output matches **Hugging Face**, **bit-identical** whether batched, cached, preempted or split across workers.
- Fault-tolerant**: killed, frozen or stalled workers are caught by heartbeat and progress watchdogs; their requests **resume elsewhere with identical output** (384/384 through **60 SIGKILLs**; no request lost in **300** fault-injection seeds).

tandem — Python, C++ and CUDA, no torch.distributed or NCCL: distributed training from scratch

- Built **DDP**, **ZeRO-1/2/3** (FSDP-style sharding), **GPipe** and **1F1B** on its own **ring all-reduce**; a **C++/CUDA engine** (lock-free shared-memory rings, CUDA IPC, a fused reduce kernel) runs all-reduce **1.6–2.8× faster** than its Python engine on a **T4**.
- On **2× T4**, **DDP** and **ZeRO-1/2/3** train **bit-identically** to **PyTorch DDP/FSDP** on **NCCL**; all-reduce **within 2% of NCCL's bandwidth** at 64–256 MB; on an 86M-parameter GPT, **ZeRO-3** cuts per-GPU memory **17%**, **1F1B 32–52%** vs **GPipe**.

quorumbd — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: **LSM storage**, **Raft**, a self-rebalancing **multi-Raft** store, **serializable ACID transactions** (Percolator), a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark.
- Proved by **deterministic simulation** (**10,000 Raft clusters**, **610K crashes**, histories machine-checked) and a **TLA+** model (**337K states**); found **6 real bugs**, catches all **15 planted**; **time-travel debugger** rewinds runs by replay.

Also built: tessel · faultline · kiln · ferrolm

- tessel**, a tile language for GPUs and TPUs: GEMM at **parity with cuBLAS** on a T4, attention **1.3–1.9× faster than PyTorch SDPA**; **kiln**, an ML compiler **faster than torch.compile**, **ONNX Runtime** and **JAX/XLA** on CPU.
- faultline**, Jepsen-style fault injection that tested **4 etcd release lines** (**1.26M** operations under **288** faults); **ferrolm**, an LLM server with a paged KV cache and continuous batching (**3.4× throughput**).

OPEN SOURCE

- PyTorch #199441** (under review): checked all **37** Inductor-skipped tests in `test_torch.py` on **CPU** and **2× T4 GPUs** and removed **13 stale torch.compile skips** whose cited bug was fixed in 2024.
- PyTorch #199646** (under review): traced **19** skipped determinism tests to a forward/backward **mode mismatch**; **8** now run under `torch.compile`, **11** stay skipped with verified causes (e.g. **bit-identical** gradients, no atomic adds).

EXPERIENCE

Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India · Early-stage startup (incorporation in progress) · Sole engineer across every product

- Sole engineer** on **Waggle**, a delivery platform of **three Android apps** (riders, shops, customers) from one **TypeScript / React / Capacitor** codebase on **Supabase PostgreSQL**: **53 tables**, **90 RLS policies**, **154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** (45-second offer window), **4-digit handover codes**, **Razorpay/UPI** payments reconciled by **UTR**, **GST invoices** and **pg_cron** billing runs.
- Security-tested** it with **150 automated attacks** on its database access rules (all blocked) and a **42-step Playwright** run.
- Built **Populace**, an open-source multi-user testing tool (**Node.js**, an **npm CLI**, **GitHub Action** and **Electron** app); its **first run** found **5 defects in 3.5 minutes** after manual testing had passed, one breaking **every signup**.

SKILLS

CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks

Languages: C++, Rust, Python, Swift (C++ interop), TypeScript, JavaScript, SQL (PL/pgSQL), Java

Backend & Data: PostgreSQL, Supabase, Row-Level Security, `pg_cron`, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

Frontend & Mobile: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

Systems & performance: CUDA (tensor cores), LLM serving (disaggregated prefill/decode, KV-cache transfer), DDP/ZeRO, SIMD

Testing, DevOps & Cloud: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata

Sep 2022 – Apr 2026